Epidermal Growth Factor Receptor Activation of the MAPK Pathway in Carcinoma Progression

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Page 1 — Abstract & Introduction

The epidermal growth factor receptor (EGFR) plays a central regulatory role in cellular proliferation and oncogenic transformation. Binding of extracellular epidermal growth factor activates EGFR through receptor dimerization. We investigate the sequential downstream signal transduction across cellular compartments.

Page 2 — Methods & Early Signaling

Phosphorylated EGFR recruits the adaptor protein GRB2 via its SH2 domain. Subsequently, GRB2 binds the guanine nucleotide exchange factor SOS1 to form a stable binary signaling complex at the cell membrane.

Page 3 — Intermediate Cascade

Membrane-anchored SOS1 stimulates RAS by catalyzing GDP-GTP exchange. Active GTP-bound RAS directly binds and activates RAF serine/threonine kinase, initiating the cytoplasmic kinase cascade.

Page 4 — Results & Kinase Phosphorylation

In our experimental assays, active RAF phosphorylates MEK at dual serine residues Ser217 and Ser221. Activated MEK directly phosphorylates ERK to elicit full enzymatic activation of MAPK.

Page 5 — Discussion & Conclusions

Our findings demonstrate that active ERK translocates into the nucleus to stimulate transcription factors. Consequently, EGFR signaling drives cellular proliferation through sequential MAPK activation. Inhibition of MEK failed to alter EGFR phosphorylation but completely blocked ERK activation.